

"Emotion Cognition Interactions As Deep Active Inference" — Ryan Smith

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[Music] so yeah I mean the so this talk is gonna be primarily a kind of like theoretical you know in silico active inference kind of talk I'll try to sprinkle in some empirical data here in there but but so yeah my um so you know one of my major interests is in more or less ways of understanding conscious and unconscious emotional processes in the context of emotional disorders and specifically I've been really interested in seeing if you can find ways to you know understand these sorts of processes add a computational level of description and so I'll tell you about one particular type of emotion cognition interaction that I've done a fair amount of work on and recently have tried to develop deep act of inference models of so and the main the main kind of goal with this and this the specific type of emotion cognition interaction that I'll tell you about is really to an individual difference variable called trade emotional awareness and and the idea is to use it as just an example of sort of a relevant trade trade psychological difference that's sort of clinically relevant where the idea is is that hopefully even though you have this sort of single phenotype of low emotional awareness that's just something clinically relevant and that doesn't mean that there's a one-to-one mapping between that phenotype and the underlying mechanisms that are causing it and so the hope is to use both neural imaging measures and computational modeling to sort of figure out okay what are all the different possible mechanisms that could be causing this can we use that to figure out ways to identify which mechanisms are operating in which people and then use that to inform treatment selection on an individual basis and potentially the treatment development so that's that's the kind of general aim I have with using emotional as an example so first I'll kind of give you some background context what is trade emotional awareness and a little bit about its clinical relevance and I'll then give you a crash course on what we've called the three process model of emotional episodes this is a very kind of kind of like broad way of dividing up different processes they contribute to what happens when you have an emotional experience and that

we have more or less come up with as a way to kind of tease apart what the different places are in that process of having an emotional experience where things can kind of be different between different people and that could explain differences in what happens in your body what happens when you feel what happened in your body how you interpret it how you use it to make decisions things like that and then I'll tell you about the kind of active inference version of this that I put together at UCL and then science and what if anything that might have to add in terms of understanding possible mechanisms okay so first thing emotional awareness and it is it so some people it's just kind of this common clinical observation that some people understand their emotions or have trouble they some people a lot more trouble recognizing and understanding their emotions than other people do and at a low level this can lead to things like mistaking negative emotions for signs of sickness clues the kind of thing like somatization or somatoform disorders and some people kind of divide up emotion space in a really coarse brain way like they just kind of feel bad or good but yeah that's about it which is it all that action getting right if I know I'm sad I have a much better sense of what the possible causes of what the likely causes are what I got to do to regulate my emotions etc so bad versus good isn't all that informative or action guiding and it also for the same reason can lead to poor interpersonal problem-solving you know if I'm not good at recognizing my own emotions cry also am I good at understanding other people's right and so there's a lot of ways that this kind of plausibly has an influence and socially and in terms of regulation for yourself and so unsurprisingly almost also I think psychotherapeutic approaches aim to improve emotional awareness in one way or another so the main sort of instruments that we've used to measure emotional awareness is something called the levels of emotional are in a scale and it's technically performance based so it's based on self-report but it's performance based in the sense that it's not based on what you believe in yourself report and basically what it does is it just asks you to describe how you'd feel in like 20 different hypothetical situations and then it just takes the sorts of terms you use and then just scores them based on their kind of level of specificity more or less so like if you say you're sick that'd be like one point it's like a ploy somatic kind of thing if you just say bad or good there'd be two points use a single emotion word like sadness or anger the three points if you can talk about say feeling multiple things at once before so that's the main way that people sort of objectively or performance based kind of get a sense of what cognitive the wet way what sort of concepts you actually use to understand your experience spontaneously when you're thinking about it and and there's a pretty sort of large literature on this nowadays so lower level s4's have been found in quite a

large number of different emotional disorders as well as I'm just as democratical disorders and even things like is such a hypertension and or irritable bowel syndrome and higher scores are associated with greater empathy emotion recognition openness to experience lower impulsivity it's a lots of things like this and it's because of that reason in part it is part of the RDoc matrix so largely by just with motivating its importance and so then i mean the idea is like i mentioned before and it would be helpful if you want to know what the cause of this is in different people to have some understanding of what the underlying cognitive a neural processes are and really the only way so our attempt to try to do that was to use this kind of broad three process model the thing that will tell you about and so I can't really go through all the kind of supportive evidence to back everything I'm gonna say up so but we've published a number of a series of reviews kind of covering this and updating it in various ways but I'm sad I was going to show you a fun video imagine that you were watching a car really serenely going down the road and then like a zombie jumping up at the screen and screaming at you you should have all jumped at which point I would have said okay what just happened right and I would have used that to guide you through the process model of an emotion episode so you missed out but so this is in the kind of box ology depiction with the three process model is the more or less you start out with some sort of event it could be real it could be remembered it can be imagined but you're representing some event right in some way and then once it's represented then you both implicitly and explicitly evaluate along a bunch of different dimensions this can be really a little level like condition response kind of thing but it also could be higher level like you know is this something controllable is this congruent with my goals and you know things like things like that and so that is what we call the initiator effective response generation processes the idea here is based on that evaluation you have some kind of allostatic prediction right about what sorts of cognitive and metabolic resources you're gonna have to bring to the table in order to deal with it and bring yourself sort of back to preferred States and that responds broadly again has two different kind of components to it has a cognitive component or a central component and then AM peripheral physiological component so the end the cognitive component is going to be things like initiating attention those interpretation biases motivations to say approachable anything with that whereas the involuntary bodily response the peripheral component it's going to be things like automatic changes in facial expression automatic changes in heart rate muscle tension things like that and so once that all kind of gets going and I should say this is all kind of going at every moment so it's all feeding back influence again as well but then you kind of have some update

in your model what it does it just happen right what what what just changed in my body and how do I interpret it so and I'll go into more detail on that but this is me where you'd say okay I feel an increase in heart rate I interpret that as indicating that I'm afraid right something like that and then finally it's important to realize that at least briefly there's evidence to suggest that all of these things can be briefly represented unconsciously without you being aware of it so there is this sort of further process that you need where these representations are say are selected and sort of maintained active in some way so you can kind of hold in working memory and use them to make the deliberative decision we've just called that conscious access working memory you know access to working memory something like that and so yeah as a fear example had you seen the jump-scare you know it would have been like event screams zombie you know the evaluation unexpected potential danger you know some low-level association between loud scream and danger and then you would have had some change in your body that you could sort of represent that way this is actually data based on you have a bunch of people kind of draw they feel on their body when they say they're feeling different emotions and they there are these kind of like interesting reliable perceived patterns and so you have something like that I'm gonna have the initiation of some kind of threat bias and some avoidance motivation you know you'd feedback and you'd represent those bodily sensations you'd identify as corresponding to your learned concept of fear and then if that all gets accessible you'd be able to say I just got scared because of the abrupt unexpected job together right so that's kind of so process one generate the effect of respond process to represent it and this kind of hierarchical way and process three gain access to it hold in working memory to use it and I can I can do this for other things right so it could be you got denied a promotion that was goal incongruent and you direct the blame at other people say your boss great you have this anger kind of like bodily feeling right it's really active negativity bias but an approach motivation right you represent the and that you think you're angry and then that becomes accessible and you could say I'm really angry at my boss because he didn't give me the promotion and I deserved it right so hopefully that gives you guys kind of the intuition and if you kind of zoom in on this internal state representation part that's gonna be a big chunk of the modeling then you can kind of think about it is again that kind of low-level you you have this feedback from your body and you representing something like a valence valence interoceptive representation along with some semadar motor stuff like facial expression changes things like that but then you can have to map that in some way up to some higher-level interpretation and that's going to depend on what you've learned right like

emotion concepts for example are different in different cultures so it's not like there's one kind of true answer about what of the bajillion different multidimensional by states it could be in and some particular emotion concept but you make sense of it in some way you know that you've learned and there can be differences in the granularity your specificity here action guiding this right over the ways you come to carve it off and so this could again be you feel a body feeling like that you map it on to sadness with priors that come from the cognitive part like unexpectedness loss uncontrollable and this is actually meant to be like fully sort of general and analogous to say like a visual case forgets to say a visual image like that you'd map it on to the concepts trouble right in the context of other aspects that you're representing say like tools shed and i should say that this is like a huge simplification right there's definitely not two levels of representation in body states if you're interested in this we've published a paper a couple of years ago laying out based on a lot of kind of just nitty-gritty about anatomical and physiological responses Integra characterizations that you can kind of lay out and even this is oversimplified 8-level kind of hierarchy for how you represent interceptive States and kind of integrate them with higher-level information from other senses in memory and but good enough for our purposes here and and then the final process and so we've kind of based this largely on this work on what are called global workspace models of consciousness or at least the empirical support for them which broadly shows again that say if you play somebody a sound but you they don't hear it because it's distracted by something you get this kind of local representation in auditory cortex but only when it's detected does it have this kind of global broad influence on Association cortices throughout the brain see ideas consciousness requires accessibility of the information carried by that local representation via this broad sort of influence updating a model more broadly you can think about it that way and so here the idea would be something like you can represent you could have some representation of a body state like that you could have primed a concept like sadness and you could have primed you know things like unexpectedness loss etc but that may or may not you know become broadly accessible and if it is then you could say things like I'm sad because I just lost a friend and I don't have much energy right now but maybe only the body state becomes accessible then you be able to say I feel really a little energy but I don't know why you know maybe I'm sick or you could have access even to the interpretation but not to the underlying sort of causes right and then you could say and this happens fairly often right I feel where it's out of I'm not sure why there's in real life there's a kind of top-down process in the sense that you're in a context and there are norms and expectations in what you might feel in

that so those schemas are sort of available as a level priors yeah and you could you can think about that as as priors coming in part from this kind of cognitive appraisal dimension sorts of things but that could also act as priors on just what emotion concept you you think best fits the I guess I'm thinking of it even as prior to having any bodily feeling and as to priming that in some ways and maybe even increasing the likelihood of having certain or even selecting out of sort of white noise bodily feelings the possibly of labeling something that doesn't have a strong valence attached to it but assimilating it into that expectation yeah no I totally agree I mean there's questions about whether that's coming in at the response generation part or after great I mean either you're not generating that response to begin with in which case you're not gonna have a bodily feeling because the body doesn't do anything right or it could be the way interpreted after the fact right so I mean it seems like this is where you can kind of disentangle a couple different things potentially so I don't think I probably understand nothing I mean nothing I've been talking about I think is that the level low enough to be something like painting yeah I mean there is I mean there is there's a lot of work on pain or presentation actually within that like koolhaas workspace you know sort of approach to conscious processing and there I mean depending on the reason or the actual basically their circumstances in which the pain representation probably gets a lot higher up like into the insula for example but you still don't become it doesn't have to be but it wouldn't have to be conscious that could that representation can happen unconscious I mean we but so anyway so I mean that's that's kind of just a brief kind of crash course and just just to kind of frame a few of the kind of neuroimaging results that all kind of you know sprinkle in throughout the way and I am we have kind of proposed and based in part on some of our research a kind of tentative um kind of large-scale model for the way to think about these processes in the brain and based mainly on work on the function of large-scale brain networks the idea this is that I mean you know there's still kind of up in the air exactly the right way to interpret these things but while you can't really say conditional on knowing that a particular brain area activated you can't really know that much about psychological what psychological function was going looks like you can actually do this quite a bit better if you know what brain areas you're talking to one another there's no connectivity and so based on that kind of thing you know people have kind of looked at a number of studies and proposed and for instance this paper by Lisa Feldman baronage a StopIt a ways of understanding kind of broad domain general functions of these different networks and you know one for example this thing though in cream here the limbic network looks like it's pretty strongly involved in representing and regulating this promoter

states and so it's kind of a prime target for generating these sorts of effective bodily responses and response to evaluations whereas the somatic motor Network here and salient salience network components especially sort of insula and anterior cingulate and look like some of that is also involved in the generation process but also looks like it's pretty important for representing interoceptive and somatosensory states to be a lot of the kind of body state representation stuff and the default mode network thus can come here is pretty strongly implicated in a lot of conceptualization related processes right so multimodal integration trying to make sense of things as a semantic level I'm watching these sort of large scale simulations we're kind of like don't along temporal scales based on and drawing from one for memory things like that and then the yellow thing executive control Network largely implicating things like working memory control and it's a big chunk of what activates during conscious versus unconscious contrast and so the kind of thing we have in mind is you know you have some perceptual representation it either gets interpreted you know via default Network processes or directly has some sort of conditioned Association thing with limbic Network processes that ends up generating a change in body state and some modular Tori effects on cognition likely based on influencing these sort of broaden rule module Ettore patterns from brain stem or modulators that be kind of the you know tweaking precision here and there kind of a thing and body state information then feeds back in and updates your model and so representing your body in a different state now that then can feed back and further get interpreted and conceptualized as an emotion or as something else and via the same sort of default Network processes all that stuff can then compete for maintenance and working memory and then all of it can both consciously and unconsciously then influence actions life in some way right so again very very broad we do have some studies consistent with it but I wouldn't say it's you know at all kind of you know take you to the bank just kind of something guide hypotheses and so you know we've thought ok well you could have individual differences in the response generation stuff in the representation stuff or an accessibility and those could each you could tweak each of those independently and produce a similar kind of phenotype right so different things could be going on in different people and knowing what's going on and what person would be important potentially in terms of their veteran sorry would you classify facial expression and display as part of a response because because there are big cultural differences and sort of training for the ability to issue display yeah absolutely and there can be some kind of you know top-down control stuff to beret where you could have the automatic tendency to do one thing but then suppress it give them some norm right and so that

also it gets pretty complicated best yes here on the course very general level without even getting into cultural differences to get a sense of the general picture it seems to me maybe we need to talk about community meaning is obvious culturally framed so experiencing a deep emotion for humans has something to do with the attribution of meaning funding like a narrative the cost related cause and then say the struming to receptive surprise it probably table go back you know yeah absolutely no I don't disagree at all I mean I and I mean not you know I'd like to pack some of that into you know the sort of you know like the hierarchically expanded version of this like a value of appraisal sort of thing I'm talking about right where you can definitely have one of these top-down influences even when they're not culturally specific right like amygdala they make a little respond pretty strongly to images of food when someone's hungry but not when they're full for example right so I'm going to sort of top-down an expectation about what's relevant you know in your situation be very context specific right so I mean there's definitely a lot of that and a lot of the way that you evaluate things in terms of say consistency with norms and values and consistency with your current goals all of that is going to have very much a sort of top-down component and influence the response you have to whatever you think is going on but it will also influence other things to you right like the way you interpret what state you're in half of the facts yeah I just totally agree okay so like I mentioned we've done sort of a number of studies over the last few years testing various aspects of this and finding neuroimaging results both structural and functional in we at least think are consistent with it like I said I mentioned some of them along the way but and all of this is primarily to motivate why it's worth trying to do a fancy active inference model version so since we haven't really talked about active inference at all at least not the Markov decision process for me and I could probably be worth giving people kind of a quick walk through and you know if you've seen this kind of Bayes net depiction before can be kind of intimidating it's a billion variables you know doesn't say it doesn't say anywhere what they mean and so I'll try to give you kind of a kind of step by step here real briefly if you get rid of most of that and you just have this oath thing at the bottom this connection up that we'll call a and then this s thing which is the state view the state you believe you're in and then D which is essentially your prior expectations about states then this is just a pretty simple sort of depiction of kind of one-step Bayesian inference you have some prior expectation you get some observation you infer what that observation means based in part on what you expected ahead of time right and the Saudi is your priors the a matrix essentially just describes your beliefs about the relationships between States and observations right like something white and circular is more

consistent with baseball than football right I mean that's basically all it all it amounts to right so then if you're going to to expand on that then you know we don't just observe things statically right so you need a temporal component to this and that's where this B matrix thing comes in so that's essentially your beliefs about how things how states evolve over time so once you figure out what you believe about what you're in about where you were what state you were in at time 1 this B matrix lets you gives you a prior and ok what a state do I expect nowt to be in at time 2 and so then that becomes your prior you get an observation at the next time point and then you infer a posterior over States at time 2 which can then also feedback and update your beliefs about what happened at time 1 but but no central all you're doing is just making this thing temporarily extend it so beliefs at one time quite act as priors for the next time plane but this is still fully perceptual and so then to add an action model on top of it and you have to add these things so PI here is just your set of allowable policies choices you take yourself to have and this G thing is the expected free energy of each policy essentially Co encodes your something like the estimated value you assigned each policy and and that is evaluated with respect to this c-vector thing which encodes more or less what you want and what you don't want so an active inactive inference Linga what's your prior preference distribution and and then finally so okay just give you a sense so GE the expected free energy that is one way of displaying the equation I won't go into it in detail but the thing to understand more or less is it has a component that involves minimizing the deviation between the outcomes that you think are going to happen if you choose a certain policy and the and your preferred outcomes see secondly you're minimizing the public label divergence between the outcomes you think are gonna happen and the outcomes you want give me a choice you could make and then the second component and involves a way of also basically policies that are having higher value if you think that them leading you to transition to certain states it's going to be really informative it's going to minimize some certainty more about what states are actually so yeah so I mean all you should take this to mean is just that the the value of policies so that the posterior distribution over PI is going to be based on the free energy calculated with respect the expected free energy calculate the perspex you see so so more or less just this just means policies have higher values if they're gonna get you closest to what you want and if they're also going to give you more information about where you know what's actually the case because a lot of times before you can know what you should do to get what you want you have to know enough about you know what the state of affairs actually is and so then finally the last step and is adding these additional things and which won't matter a ton for what I'll

show you but more or less these additional kind of hyper parameters allow you to allow the model to maintain estimates it's essentially its own confidence in its actor model itself so this sort of beta-gamma thing is more or less a way of having a prior and then updating this expected precision term which more or less reflects your confidence in how good your action model is and that ends up actually trading off influence with this thing which is essentially a fixed form prior over policies so it's sort of the policies you expect a priori to select over others take thing we're it's not like that just like habits more or less and so the way this ends up working is is that policy selection will be more driven by habits if gamma is low if you have very little evidence there but very little confidence in your action and having a prior prior for high beta is going to buy us your initial gamma so it toward being lower which means you're going to favor acting based on habits so that's the way of interpreting this thing and and so that that will happen kind of just on its own I mean that bull just falls out of the expected for energy equation right so so you just have the epistemic component is that latter component and the pragmatic component is the the earlier part of the right side so so for instance like it just it just will happen automatically that whatever value whatever whichever policy has the lowest expected free energy will will be a trade-off between those two there's nothing waiting than outside of just so an empirical sound like them so if you want to like fit behavior empirically you know I try to do this before where you can actually throw in an additional throw in an additional term it was actually wait you have a stem a component versus the pragmatic component but in practice what will happen is because the because the M the C vector because the the magnitudes preferences can vary continuously then just the higher the magnitude is the the greater the weight the pragmatic component will have so having a low magnitude C just automatically makes it so you're more driven by this nice value like you have production to see so maybe if you're hungry then you start being temporary yeah it's gonna say yeah that's that's not built into the standard NDP formulation but yes I mean there's there's definitely fancier things you could do Casper has tried to do some of them well stay tuned about so so anyway and just briefly I mean you can't really see this a lot well but and the active inference formulation I'm just showing you also has this kind of proposed accompanying neural process theory it allows you to essentially simulate um plausible neural activation the you want to see given the dynamics in the model it's not gonna be that important for what I'll show you but more or less and you just end up and you end up being able to derive these update equations that just cast the essentially the equations for updating free energy and for excitement free energy in terms of these two different sorts of prediction errors

called state prediction errors and outcome prediction errors and then those then those errors then feed into calculating or updating the free energy and expected for energy estimates and then you can just do used agreed units that creative dissent on those and what ends up happening is that belief updates for a moment moment correspond to changes in them activation levels and different of these kind of simulated neurons whereas the actual matrix entries like they the major can matrix entries in a and B and C in D that indicate more your kind of like more stable beliefs about relationships between things those end up corresponding to synaptic connection strengths which you can learn but you update them more slowly across trials but like I said it won't matter that much so applying that them to understand emotional awareness and is something that we've done for you and and one thing so you can just ignore this top part basically all of it all that adding the second level on means is that you just attack the same thing on a top what I just showed you so so the state your state inference at the end of each trial at the lower level just ends up being the observation that feeds into the same exact structure at the level above so for instance you'd run the lower level like three times and then that would correspond to three observations over time by the higher level and so for the kind of task we put together to model emotional awareness is this kind of emotion focus working memory task more or less the idea is is that the agent is presented with some kind of effective response at time one and has to infer what emotion it's feeling and then some time goes by its presented with another affective response has to infer what it's feeling now and then this kind of delay period where it's gonna hold on to that information and maybe later at which at which point it's exposed to a third an affective response then it has to say okay is what I'm feeling now the same or different from some of the stuff I was doing before so it's kind of a reflection on current versus past emotional experience this is actually you know thing not too dissimilar from stuff that can happen in therapy right and it's also well yeah I'll get to this so but the so the idea ultimately with with the model as I'll show you is you end up having these two nested tasks so first you have this effective response that kind of comes up from your body and you represent it and which can have you know various dimensional components valence arousal motivation your kind of interpretation of a situation things like that and the first task is to infer which emotion concept that you currently have acquired best accounts for that pattern lower level representations but at a second level you have to take that and map it into working memory then you have to kind of do something with it memory to be able to self-report it and use it in collaborative cognition so task one is just to figure out what you're feeling at a given moment and task two is kind of accumulated evidence for what different things you're feeling

over time and then use them to deliberate to make a choice and I just say this particular task is motivated by the fact we've actually done some empirical work on emotion focused working memory and with just little tasks like this where you show people emotional images and you either ask them to hold what emotion they felt in mind or just hold the visual image in mind or hold like their bodily sensations in mind or hold nothing in mind over some delay period then they have to make some you know judgment afterwards about whatever they were holding in mind and what you can find is actually consistent with the three process model I showed you you get in the emotion condition relative to just holding visual images in mind you get a lot of this medial prefrontal and anterior insula kind of default mode Network kind of stuff whereas in in both conditions actually emotion greater than rest and also envision greater than rest you get this dorsal lateral front a parietal thing do you expect for any kind of working memory so so it's kind of like the representation may be specific but what's holding it in mind is general and it's also falls on previous we're showing that just when people pay attention to their emotions and just recognize them you also get this but so okay so so the first level in the model formally so it has this kind of structure where there are a bunch of different things that you can observe you can observe different balances different arousal levels different approach or avoidance motivations and three dozen general different kind of evaluated contexts that you could be in its got like a neutral context to social threat context or a physical threat context and again this is very just kind of toy model to get across the general structure and but the idea is is that what it is to have a quiet different emotion concepts and is to have learnt particular mappings to different sorts of regularities that occur across those different lower-level representations so for instance and you can encode you encode that in for instance the a matrix here so this basically says that sadness corresponds to negative valence low arousal avoidance and social correct right whereas happy on the end corresponds to neutral wait I might I might have that wrong oh no sorry that's heart attack like that so it's sad this panic statement heart attack so happy is negative high arousal avoid and physical right so I mean that's that's kind of what it looks like this is encoded in those connections more or less and for this particular model we're just assuming that within a trial and I motion remain stable and so the B matrix is an identity matrix and and then down here but process right you'd interpret your situation a certain way it's selective stay to like increase heart rate policy right or something something more high dimensional than that obviously but and so we're not explicitly modeling that there we're just assuming a particular response has been generated that it's consistent with your evaluation of the context and that you're just being the agents kind of being presented with the

result of that response and so whichever is highlighting here is that we're kind of assuming that's the case but we're not actually modeling the response generation that that even lower level policy selection first and then the second component of the first level is there is a kind of selective attention component building and the idea here is to figure out what you're feeling you have to kind of selectively attend say okay what they'll into my feeling okay you know what does my body say feel like oh or I feel like do I feel like I should run away or you know approach you know what do i what are my actual beliefs about situation things like that this is actually the exact sort of thing that often people do in like cognitive therapy for example where you can attract patterns right and like what you're thinking what you're feeling what you did you know stuff like that to try to increase people's awareness so it's meant to kind of mimic that process and so the idea is is that the agent has to kind of choose what to pay attention to across the trial to infer what emotion makes the most yeah best fits the experience and so you can kind of the way that gets implemented is that the actual the actual probability distribution in the a matrix is becomes higher dimensional essentially it's different conditional attentional that you're in so if you're in the attentive valence state then the a matrix looks like that if it's a tend to arousal looks like that but to tend to motivation it looks like that and if it's attend to context it looks like that right so that's that's the way it ends up being structured and so so for the first level then you can use that all by itself without tacking on the second level at all to do this kind of single single emotion recognition process right the emotional state representation component of the three process model and and this is what it looks like when the model performs successfully so here the agent was presented with an effective response that was most consistent with sadness according to the concept it had learned and so you can see what happened as it first chooses to pay attention to context and then it chooses to pay attention to arousal and then at that point it says okay I'm confident based on the social threat context and low arousal that sadness is the right so report status and then you can just kind of see over here how the confidence kind of shifts slowly toward being confident most confident in sadness by the third time going to trial there on the right and this stuff on the bottom is just kind of simulated and simulated like regulated potentials that you would see but don't worry about that so then the the second level then and the working memory component involves doing a sort of similar thing so basically what you have here is is now once the agents inferred what emotion they sad or panic or what other bodily states sick or heart attack you know however it's categorizing its data at a given time those provide evidence over time for different things that could be held in working memory so this could be

just being sad just feeling panic feeling both sadness and panic etc etc for different combinations and then you have another independent hidden state factor that corresponds to the comparison state so essentially what you find out at the end the third emotion that you're feeling you have to judge whether it's the same or different than one of the original two and so the a matrix for that one just looks like that and which I can go into if you want but it just describes that mapping I just described and then there are a couple additional hidden state factor you have to build in here like you have to build in beliefs about the task structure the agent has to know for instance that it's at time point one you know in the working memory task time point to whether it's in the delay period etc and then it also has to choose to report at the end of the trial just two things just same or different you know feeling the same as before we're not and that leads to observations where it gets some feedback were either it was you know it gets it right or it gets it wrong and the C matrix here just basically says it wants to get it right it does want to get it wrong so this part would be modeling the kind of conscious access process more where once you have these representation it's you map them in and hold them in mind you can do something with them and so a successful trial with the model here looks like this so at so the first time point so the this middle panel here corresponds to the single state inference at the lower level so first it infers nothing so we just call that blank the second time Boyd and at first it was feeling sadness that the third time panic then blank again because it's the delay period then at the fourth time point it thinks it felt neutral and then blank again that's just the structure of trial and what what it experienced at each moment and you can see at the second level what happens is it slowly builds a confidence over time that the sadness plus panic state is the right one right and then it holds that information in mind over this delay period after it observes a second emotion and then uses that at the end to choose to say different way in which case it observes that it was correct what do you say and so I mean it is these are beliefs about how how it feels changed over time right well I'm except for instance say say you had a panic I mean this is uncommon you have a panic attack and then you're really sad the rostral day that you had a panic attack so so that the the potential usefulness of this kind of thing other than just showing you know fancy seeming flashy simulations is that you know it allows you to kind of interrogate what are the different ways that things can go wrong what are the different possible mechanisms by which things could break down okay and where you could have different potential intervention targets so the first one and all this sort of mentioned this really briefly is just you can maybe you're just failing to generate responses at all in which case just means visceral policies aren't being selected in the first

place there's nothing to explicitly simulate there but I mention it because it's probably not unreasonable thing thing happens in some cases so for instance in one study we did we actually did find that the thinner viscera motor cortices were the lower emotional awareness towards work so suggestive more interestingly we've done some we reported in these case studies these extreme sort of rare cases where and for instance this woman just reports she's never experienced an emotion her whole life just flat just emotional this experience and she's really like paradoxically distressed by head because she ultimately committed suicide because she said her life felt meaningless without any emotion and when we did like when we did like skin conductance responses on her totally flat no responses to emotional images whatsoever but but it was more just like they were kind of incoherent responses because she would be talking in a therapy session about like for instance her marriage feeling and tears of like well up in her eyes but her face would be totally flat otherwise and like therapist would say hey it looks like you're like crying she like she like yeah you know I don't control it as my eyes just do that sometimes like and so point being it's not inconsistent that this is the explanation in some cases and I mean there is some actual developmental literature you know suggesting ways this could kind of happen too right so if you look at and the sort of drawing version of how people's body feels when they're feeling different emotions six-year-old it's like if you look at sadness six-year-olds it's pretty like nondescript it's basically nothing whereas in adults it's really highly specific right so you could imagine ways in which the six year old version just persists under certain sorts of like unfortunate developmental so the next version so just like all failure to infer and this is this corresponds to something we've talked about in the past called affective Ignatia the idea being something a lot like a visual agnosia where yeah okay right so the basic idea is kind of just like with a lake and associative visual agnosia where you could have like a an experience sexual experience like that trouble right person could drop for you and everything but you ask them what is it they have no idea right or they miss recognize it as something else sorry ghee there's a similar sort of thing that could happen but in these sort of emotional realm effect of ik know Jia where a person perceives and could describe the bodily feeling but just doesn't recognize what it means and and often what will happen in this case is that it'll be miss recognized with something else and so in this matters ation case you might say think you're feeling you know have a bodily sensation consistent with fear or panic but instead infer of this as a heart attack which often happens people with panic attacks often run to the ER right and so so one way to make sense of this or potentially right would be if a person's just learned to have a really strong prior expectation for threatening somatic states

and one interesting example of this is people with high anxiety sensitivity scores like actually characterizes just that and and this is actually something in part that we talked about in a recent paper we're trying to kind of give explanations for why early adversity it's actually associated with high levels of chronic oh I don't have time to get into it but if you're interested and so here I mean it's interesting what actually happens if you give the thing higher prior expectations for sickness and heart attack right for this threatening somatic states then what happens is basically the model this jumps to conclusions too quickly like all it does is pay attention to its arousal level sees high arousal and sight already know you know so it just says it having a heart attack it doesn't it's too confident right so it doesn't pay attention to the context or anything else that would allow it to say okay now I'm probably just having some anxiety yes so I'm not going to finish it at all if I don't take power through here but but at the end and so again I should probably skip some of the empirical work here just given time but basically we have shown that if you look at people with different emotional awareness levels the default mode network actually only activates more than some representation systems if you have high emotional awareness so so it looks like you don't really differentiate emotions from bodily sensations at the low end which is kind of consistent with this idea and so a third one so could be that you just haven't acquired emotion concepts right so consider like a person who grows up in like an environment with a lot of like just early parental neglect you don't have the attunement you don't have the you know social feedback signals to learn about these things and so you just don't acquire them um and this is probably also something that contributes a lot to if you want to understand things like the relationship between social support and disease we've talked about in the sort of computational variant of the trying to understand the biopsychosocial model in medicine and we've talked about it this a little bit there but and you yeah like if you don't have the early support that you're not gonna learn the things your needle you're gonna need to learn and that can lead to a bunch of things that can promote systemic disease later in life and but so here um really all you have to do to mimic not having acquired emotion concepts is just to get rid or flatten out the you know connections between the emotion concepts and what they predict right if those days don't predict anything specific that's the same thing as saying you don't have those concepts and so if you do that then let's get that and then what happens is unsurprisingly and the thing never really becomes very confident no matter what it experiences at the end of the day it's like okay I have to report something so it just reports that it feels bad it's not confident knows the feeling an emotion but it's not confident enough to pick a specific one and it doesn't want to be wrong so just as bad

and and interestingly this version predicts you'd have much longer reaction times right so in the first version if somebody has a low emotional awareness score they have a really fast reaction time because they dump the conclusions really quickly right whereas this person sits around forever and along I don't know what's going on right so like you could actually test just based on reaction times which one of these sorts of mechanisms with most possible any give them individual let's just say I think that's a very important distinction and finding because you could also make the argument that a distinction any good and bad is just a very crude and low-level distinction that should be made very quickly right reptiles could decide isn't good or bad right but this is saying no actually it's really you are it could be at least in certain people yeah be able to say you have fast reaction times you have solar reaction times therefore a treat you or even a for you a treatment be and again you know just briefly I mean we've done some work like with resting state functional connectivity and showing that you know if you think about functional connections as reflecting differentiated say like synaptic you know connection patterns of correspondent learning and then what we found is in both the default Network and in salience Network body see representation regions you have much lower so at least something about the fidelity of that kind of processing is isn't is not as great this is just some of the data and correlations but but I should just mention we have another paper that's under review right now where we actually simulate the emotion concept learning process itself with active inference and that also is kind of a cool story about when learning can go wrong and but not going to talk about that here and so then a fourth possible mechanism is just having learned by its patterns of attention this is another thing that comes in a lot actually when thinking about chronic pain in the context of chronic negative affective states and but and we've talked about before but here I mean it's pretty simple you just you just use the e matrix thing the prior prior ever policies and you just say this person has a really strong habit of only paying attention to context right for an external attention bias or this person has a really strong prior that they're going to attend only to the internal stuff right and I mean kind of you know as it's set up you have to integrate context information with the bodies didn't for me to be good at inferring you know what you're feeling or understanding it and so if you simulate these different sorts of attentional biases you get low emotional awareness behavior but they have different patterns you know in one if it's an external bias but they're kind of like averse to paying attention though their body altogether right then they just end up reporting you know patterns good whereas if they ignore context and this hyper focus on their body then they just tend to kind of report kind of like an emotion thing or a body thing kind of half and half right

they're just kind of inconsistent and the fifth one again I'm just kind of rushing through these quickly because we're short on time and something that's interesting in the context of borderline personality disorder is this idea about beliefs and emotional instability or high emotional volatility if you just believe that you could be sad in one moment happy in the next and right if you don't think emotions absolute minimal amount of stability then it actually becomes really hard to learn about them or infer them if you have to selectively integrate information about them over time right so in this case you just flatten out the B matrix down here to say look like emotions are allowed to transition to other emotions or to somatic states within trial and then what you end up with something like this where the things just not confident at every moment it just rechecks the same thing I looked at before because then something could have changed and when you end up with this thing's like it starts out believing that it was feeling panic but by the end it self reports that is feeling a heart attack you know it's like I mean it might have started out as just panic but now you know now it's a heart attack so it's another thing kind of interesting and and then that the last two mechanisms are when you have to invoke front of his higher level right the working memory level and here coming again just review papers on stuff if you're interested in more detail but the idea is how do you kind of model this presence or absence of this strong connection on causal influence that actually updates the higher level and this is also motivated by some work we've done and some graphic theoretical Alice's that we've done showing that actually if you look at individual difference is an emotional awareness they actually correlate with lower network density and with longer average path lengths so so those are just two measures of essentially a long range information integration so if you have higher emotional awareness you're better at propagating the signals over long ranges right so kind of consistent with this general idea this is a positive thing and here it's a similar kind of trick you just take these connections from the first level of the second level and any of them that involve updating based on emotional information at the first level you flatten out which is essentially the same thing as saying the first level representations no longer have causal influence on the higher level right and if you do that then what's really cool is you get something a lot like unconscious emotion priming right so the first level it is representing sadness and then panic and then neutral right that's there but just in kind of this prime state that could have some kind of semantic priming effects but it doesn't update the higher level at all so the thing can't hold the emotions in mind work with them at all and it gets it wrong when it needs to report something or think about supposedly I think about what it's feeling and then the final one and one that's kind of interesting in clinical

context with some of the clinical clinicians that I've worked with in the past has to do with this idea about sort of state differences in reflective capacity the idea here is is that again the image doesn't show up that well but there's this really like well-known result that cognitive performance like the ability kind of reflect on things dying in control working memory and all that stuff tends to be best at this kind of middle level of arousal if you're like really highly anxious then all of a sudden you can't reflect on anything at all and and decision-making becomes a lot more impulsive and I reflective typically not adaptive and one way to think about this is that in high arousal states and the predictability for the precision assigned to the higher levels of processing like that reflect these sort of long time scale regularities those just kind of get assigned to a procedure because in threatening context sort of historically yeah now where you're eating right it doesn't make sense to assign high precision to those to the high levels in those contexts and so you can kind of think about this as in a state dependent way decreasing the transition precision at the higher level which Thomas Parr and other people have previously already shown this is a way to mimic lower working memory capacity as well it's essentially the same kind of idea so what happens in this case is and even say if at timepoint two it knew that it was feeling it was feeling in this case panic then that just kind of decays away over time so it can't actually even though it knew it at one point in time it can't hold it in mind and actually use it and and so that's just yet another kind of mechanism and again I wish you could see it better on the image because it it really does kind of just decay away until the probability distribution becomes flat by the end but but so anyway so that I mean the idea is just basically you can use this and you can show look there's at least these seven different independent mechanisms that could be messed up or working sub-optimally that could lead to the same or very similar clinical phenotype but they have different underlying causes and you would target them differently and we've actually enter in our paper we actually outlined and you can't see it but for each of the different mechanisms there are different sort of clinical evaluations that are currently used different relevant measures that are currently used try to you could use to evaluate them and different intervention and they look like they target some more than others and they're also gaps right where you can design things that would more effectively target one versus the other but a lot of it is kind of out there somewhere it's just not framed in this way or selected you know treemail are selected based on this kind of thing and so the basic idea is is they like I said there's a bunch of these different possible mechanisms you could actually test them based either on different predicted normal response patterns or behavioral patterns like reaction time differences and and you know if

that ends up working right you have to do a criminal experimental work to actually test that but in principle you can use that to identify mechanisms that operate in single individuals and then use that for targeted intervention so I'm with that tons of people to acknowledge but top of the list would be Carl first and Thomas par Richard Lane Maxwell Kasper and all Badcock [Applause]